

Zero Trust Architectures in Cloud Networks: Enhancing Security Across Global Enterprises

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Abstract- The growing usage of cloud computing among worldwide organizations has brought tremendous security issues, calling for a shift from the conventional perimeter-centric defenses to more effective, adaptive security models. Zero Trust Architecture (ZTA) provides an encouraging model by imposing ongoing authentication, least privilege access, and micro-segmentation to combat insider threats, unauthorized access, and data breaches. In this paper, a Zero Trust-based Security Framework (ZTSF) for cloud networks has been proposed by incorporating AI-driven identity and access management (IAM), policy-driven micro-segmentation, blockchain-secured trust mechanisms, Zero Trust Network Access (ZTNA) across multi-cloud setups, and AI-fueled threat intelligence & response. The suggested framework employs machine learning for anomaly detection, behavioral biometrics for authentication, smart contracts for compliance, and Software-Defined Perimeter (SDP) for workload isolation. It is assessed via simulation platforms like Cloud Sim or Mininet in terms of security effectiveness, latency overhead, and resource utilization. Via case studies and comparative study, we show how ZTSF enhances cloud security for businesses, making it resilient to sophisticated cyber threats and yet maintaining efficiency in operations.

Keywords: Zero Trust Architecture, Cloud Security, Identity and Access Management, Blockchain, AI-Driven Threat Detection, Micro-Segmentation.

I. INTRODUCTION

The sudden growth of cloud computing has transformed enterprise IT infrastructure with scalability, cost-effectiveness [1], and global reach. Yet, as more organizations move sensitive information and mission-critical applications to the cloud, legacy security models have failed to keep pace with advanced cyber threats. Legacy perimeter-based security designs, based on implicit trust inside the network perimeter, are susceptible to insider attacks, credential attacks, and lateral movement of attackers. Such issues call for a paradigm shift to an increasingly adaptive [2], safe model—Zero Trust Architecture (ZTA). Cloud networks operate in a very dynamic and distributed environment, such as multi-cloud and hybrid-cloud environments, third-party integration, and remote employee access. Traditional security controls like Virtual Private Networks (VPNs) and firewall-based access control rely on

the premise that within-device and within-user devices and users are somehow assumed to be implicitly trusted. But modern cyber threats, such as Advanced Persistent Threats (APTs), ransomware, and identity-based attacks [3], exploit benefit of this trust model to obtain unauthorized access. The security model of Zero Trust eliminates such issues through the elimination of implicit trust and the mandating of tight, continuous verification for all access requests. Zero Trust Architecture is based on the principle of "Never Trust, Always Verify," subjecting every user, device, and application to continuous authentication and authorization before allowing access to network resources. It is especially critical for cloud networks because data is now spread across a number of environments, and conventional security perimeters no longer exist. Zero Trust networking in cloud systems is based on a number of core principles such as identity-led security, with authentication and authorization being asserted based on end-user identities, roles, as well as on contextual conditions including device health, geolocation, and behavior [4]. It also promotes least privilege access, whereby users and applications are given the minimum privileges required to carry out their functions, minimizing the effects of possible breaches. Micro-segmentation also divides cloud workloads into insulated safe enclaves to prevent the lateral spread of threats, and continuous monitoring and adaptive security take advantage of real-time threat intelligence, anomaly detection, and behavioral analytics to detect and counter suspicious activity. Moreover, Zero Trust Network Access (ZTNA) offers access on a fine-grained policy-by-policy basis, eschewing reliance on traditional VPNs. In response to these challenges, this paper proposes a Zero Trust-based Security Framework (ZTSF) that can tighten cloud security by including advanced technologies such as artificial intelligence (AI), blockchain, and Software-Defined Perimeter (SDP). The solution consists of several critical components, including AI-based identity and access management (IAM), which uses machine learning algorithms for ongoing authentication and anomaly detection and uses behavioral biometrics and risk based authentication to mitigate credential-based attacks. Policy-based micro-segmentation also dynamically micro-segments cloud workloads and applications to restrict unauthorized access, using Software-Defined Perimeter (SDP) to create isolated security zones. To ensure data integrity and compliance,

blockchain-enforced trust mechanisms are integrated, with blockchain-based audit trails used for tamper-evident logging and smart contracts used to enforce compliance and access controls automatically. Furthermore, Zero Trust Network Access (ZTNA) for multi-cloud environments [5] enforces a single Zero Trust model for hybrid and multi-cloud infrastructure, with end-to-end encryption and continuous API-level verification. To further reinforce security, threat intelligence & response based on AI is utilized, leveraging machine learning for real-time anomaly detection and proactive threat reduction and incorporating Security Information and Event Management (SIEM) for automated security analytics.

This paper makes the following primary contributions:

- Suggested a Zero Trust-based Security Framework (ZTSF) that incorporates AI-powered authentication, blockchain-supported trust mechanisms, and micro-segmentation to enhance cloud security.
- Improved Zero Trust Network Access (ZTNA) for multi-cloud environments through end-to-end encryption, adaptive security policies, and Software-Defined Perimeter (SDP) to limit unauthorized access.
- Assessed the framework via simulations to show enhancements in threat detection, access control, and attack surface reduction for international businesses.

II. LITERATURE SURVEY

A Zero Trust Architectures (ZTA) survey of cloud security captures a complete overview of recent development and deployment in cloud security. This study integrates contributions from different researchers to scrutinize Zero Trust precepts, processes, and practices in cloud computing and information security.

A primary knowledge of Zero Trust in the Cloud by Sharma (2022) [6] comes with a focus on the requirement for constant validation, micro-segmentation, and minimum-privilege access in enterprise cloud security. The research emphasizes that conventional security frameworks are based on implicit trust, exposing cloud networks to lateral movement attacks, insider threats, and credential-based attacks. Sharma's research examines Zero Trust practical implementation approaches in favor of identity-based security and multi-factor authentication (MFA) to ensure protection against unauthorized access threats. The paper further examines policy enforcement technologies like software-defined perimeter (SDP) and adaptive access control in the prevention of insider threats.

A seminal publication by Stafford (2020) [7], NIST Special Publication 800-207, establishes Zero Trust

Architecture as a security concept that removes implicit trust from corporate settings. The standard focuses on ongoing authentication, identity assurance, and contextual access control. The NIST model brings a policy-based methodology in which the security perimeter is redefined, moving away from traditional firewall-based security towards micro-perimeters that guard individual workloads, applications, and data. The article emphasizes risk-based adaptive authentication, which real-time adapts access levels under security conditions in real time, including device health, network location, and behavior anomalies.

Research by Kolawole examines AI-powered [8] security impacting Zero Trust Architecture in cloud networks, especially on U.S. cybersecurity efforts combating foreign threats. The article features the use of machine learning algorithms in Zero Trust models to enhance real-time detection of threats and anomaly-based authentication. The study underlines the relevance of behavioral biometrics, predictive analytics, and AI-based SIEM to reinforce Zero Trust. Kolawole contends that when Zero Trust is combined with AI-based threat intelligence, organizations are able to forensically anticipate and defend themselves against cyberattacks instead of dealing with them as they occur in the wake of breaches. Federated learning, the study reveals, is one of the developing methods of securely training AI models in decentralized cloud environments.

DeCusatis et al. (2016) discuss Zero Trust cloud networks and suggest an implementation with transport access control and first-packet authentication. The paper presents [9] a novel network-layer authentication scheme in which access requests are authenticated prior to establishing network connections, such that only legitimate entities can initiate communication. This solution strengthens packet-level security, considerably limiting man-in-the-middle attacks, session hijacking, and unauthorized lateral movement across cloud infrastructures. The paper also discusses integration of software-defined security (SDS), policy-based orchestration to enforce real-time security policy on dynamic clouds.

Li, Iqbal, and Saxena (2024) [10] extend Zero Trust to Industry 4.0 and Industrial Internet of Things (IIoT), establishing a Zero Trust industrial cybersecurity framework. The paper pinpoints major security threats in smart manufacturing, industrial automation, and critical infrastructure, which have legacy systems that do not possess advanced security features. The authors propose a Zero Trust Security Fabric (ZTSF), which incorporates blockchain-based immutable logging, AI-powered access control, and network segmentation for securing IIoT environments. Their plan depends on blockchain-secured authentication to avoid unauthorized access and uses predictive analytics to identify potential threats in IIoT networks. The research also highlights the essential role of software-defined perimeters (SDP) in isolating industrial control systems (ICS) from external cyber threats. These articles cumulatively highlight

the establishment of Zero Trust Architecture (ZTA) in cloud security, AI-based cybersecurity, and industrial IoT applications.

Sharma (2022) and Stafford (2020) emphasize essential Zero Trust tenets and NIST standardized implementation, whereas Kolawole (2020) and Li et al. (2024) extend Zero Trust use to AI based security and industrial networks. DeCusatis et al. (2016) speak about network-layer Zero Trust measures, providing security at the transport level through first-packet authentication. The major research gaps are,

- Scalability within Multi-Cloud Systems - Existing Zero Trust models find it difficult to integrate seamlessly across multi-cloud infrastructures due to inconsistent security policies and un-interoperability
- Computational Overhead and Latency – Authentication and ongoing verification powered by AI create performance bottlenecks, necessitating optimization methods to reconcile security and efficiency.
- Automated Policy Adaptation – Inadequate AI-driven, adaptive security policies that adjust dynamically based on evolving threats and user behavior in real-time.

III. PROPOSED METHODOLOGY

A. Zero Trust-Based Security Framework (ZTSF) Design

The Zero Trust-Based Security Framework (ZTSF) is a fundamental departure from traditional legacy perimeter-based security architectures, emphasizing a "never trust, always verify" paradigm. Unlike conventional security architectures that rely upon implicit trust within a network of an enterprise, ZTSF demands continuous authentication, strict access control, and real-time detection of threats for all devices and users whether or not they are located anywhere. Identity-based access control is the core of ZTSF in which all actors—human user, device, or application—are untrusted by default until verified. Role-based access control and attribute-based access control govern the security policies, making permissions dynamically change based on real-time security assessments.

Mathematically, the trust score T_i for a user or device is computed as [11]:

$$T_n = \alpha A_w + \beta C_d + \gamma B_h + \delta R_t \quad (1)$$

where:

- A_a represents authentication strength (e.g. MFA verification, biometric authentication)

- C_d denotes device compliance (eg, OS patches, security configurations)
- B_h is user behavior anomaly score
- R_t indicates real-time threat intelligence data
- $\alpha, \beta, \gamma, \delta$ are weights assigned based on organizational policies

A threshold T_{th} is defined such that if $T_y < T_{th}$, access is denied or requires additional verification steps.

Through the use of Zero Trust Network Access (ZTNA) principles, ZTSF applies strict security policies at the network, application, and identity levels, ensuring end-to-end protection across multi-cloud platforms, IoT devices, and edge computing infrastructure.

B. AI-Driven Identity and Access Management (IAM)

The primary ingredient of Zero Trust security is AI driven Identity and Access Management (IAM), which it supports anomaly detection, risk-based access control, and continuous authentication. IAM solutions are used traditionally to carry static credentials that are prone to phishing and credential theft. AI enhanced IAM employs machine learning (ML) models for dynamic adjustment of the authentication mechanism as per the context and behavior of the users.

Anomaly detection is performed by a deep learning model-based model to detect suspicious login behavior. From a dataset of user login activity $X = \{x_1, x_2, \dots, x_n\}$, the model is trained with a behavioral baseline through an autoencoder neural network. The reconstruction loss L_r is computed as

$$L_r = \frac{1}{n} \sum_{i=1}^n \|x_i - s_i\|^2 \quad (2)$$

where x_i is the input behavior vector and s_i is the reconstructed output. If L_r , exceeds a predefined threshold L_{th} , an anomaly alert is triggered, prompting additional verification steps.

Another key aspect of AI-driven IAM is risk-based adaptive authentication (RBAA), which adjusts the authentication requirements based on risk scores. The risk score R , is determined using:

$$R_3 = w_1 U_a + w_2 L_t + w_3 D_r + w_4 B_n \quad (3)$$

where:

- U_a is user authentication history
- L_t represents login time anomaly
- D_r is device reputation
- B_n denotes biometric confidence score

- w_1, w_2, w_3, w_4 are dynamically adjusted weights. If R_n exceeds a defined threshold, additional authentication steps, such as OTP verification or biometric authentication, are enforced.

The combination of deep learning models and contextual risk assessment enhances security by reducing false positives while improving access control accuracy.

C. Blockchain-Enabled Trust and Audit Mechanisms

To ensure data integrity and prevent unauthorized credential modifications, blockchain technology is integrated into the Zero Trust framework. The immutable ledger records access logs, authentication events, and policy enforcement actions, ensuring transparency and accountability [12].

Each access request is recorded on the blockchain as a smart contract transaction, with the following structure:

$$TX = \{ID_u, ID_r, P_m, T_x, H(P_x)\} \quad (4)$$

where

- ID_u is the user identifier
- ID_r is the requested resource
- P_m represents permission level
- T_x denotes the computed trust score
- $H(P_x)$ is the hash of the security policy applied

Through decentralized identity management (DAD), blockchain eliminates reliance on centralized authentication providers, lessening the risk of credential-based attacks. Every identity is cryptographically signed with elliptic curve digital signatures (ECDSA) such that only authorized identities can access cloud resources.

Integrating AI-driven Identity and Access Management (IAM) into a Zero Trust framework presents several challenges. AI systems must efficiently handle large volumes of authentication requests without slowing down users. Ensuring compliance with GDPR and HIPAA adds complexity, as data privacy must be carefully managed. Many organizations still rely on outdated security systems, making AI integration a challenge. Cyber threats and biases in AI decisions can also affect trust. Additionally, AI must continuously learn and adapt to stay ahead of emerging risks. Most importantly, strong security must be balanced with ease of use, so people trust and adopt the system smoothly.

D. Micro-Segmentation and Software-Defined Perimeter (SDP)

Micro-segmentation is a core principle of Zero Trust security, ensuring that access to applications and workloads is granted on a least-privilege basis. Unlike traditional firewall segmentation, micro-segmentation enforces fine-

grained access controls at the application, user, and network levels.

The segmentation policy is formulated based on a mathematical constraint model, defined as:

$$\forall u \in U, \forall r \in R, P_{\text{amou}}(u, r) = f(T_s, R_s, C_c) \quad (5)$$

where:

- U is the set of users
- R is the set of resources
- P_{armas} is the access permission function
- T_x is the computed trust score
- R_s represents the risk score
- C_c denotes compliance constraints

A Software-Defined Perimeter (SDP) enhances security further by dynamically enforcing access-based policies based on identity-aware proxies.

A key advantage of micro-segmentation and SDP is the prevention of lateral movement attacks, where attackers attempt to pivot within an organization's network. By enforcing dynamic identity-based segmentation, the attack surface is minimized, ensuring secure access to cloud-based workloads and applications.

The proposed methodology integrates AI-authentication, blockchain-trust management, and micro-segmentation to create a Zero Trust architecture for cloud networks. Leveraging mathematical models in risk assessment, dynamic access control, and crypto authentication, the framework strengthens security, minimizes insider attacks, and enables secure cloud computing environments.

The arrival of Zero Trust Architecture (ZTA) has brought about a fundamental change in cloud security practices. This security approach does not automatically trust people or systems and replaces older perimeter-focused defenses through ongoing verification. The ever-evolving nature of cloud networks presents significant security threats including data breaches and insider attacks as well as unauthorized access and lateral movement attacks which have become more common because global businesses now standardly rely on cloud computing. Zero Trust minimizes threats through the implementation of strict access controls and continuous authentication which evaluates dynamic risk in real-time. Cloud networks depend on the "never trust, always verify" principle which demands identity verification along with device compliance and behavior checks before granting any resource access. The dynamic trust score functions as a critical component within this framework since it decides when access is allowed or blocked and when further validation is needed. The trust

score which permits access to cloud resources is computed by evaluating several factors like authentication strength, device security compliance, anomaly detection systems, and real-time threat intelligence. Artificial intelligence (AI) supports Zero Trust security by utilizing adaptive authentication methods and machine learning-based anomaly detection. Legacy identity and access management systems use fixed credentials which make them susceptible to phishing attacks and credential theft. Real-time analysis of login patterns, user actions, and risk thresholds by AI IAM systems helps detect unusual activities. Zero Trust models can detect suspicious access patterns and modify authentication requirements according to risk scores because they implement unsupervised learning models such as autoencoder neural networks. The system employs stronger authentication mechanisms like biometric authentication or multi-factor authentication (MFA) to block potential security weaknesses when it detects irregularities. Zero Trust security relies heavily on audit controls together with secure identity management through blockchain technology. There is a vulnerability in the system through which identity compromises are made possible: legacy centralized identity providers (IdPs). Decentralized identity management (DID) on blockchain allows Zero Trust security to cryptographically verify user identities while making them unchangeable and protected against identity theft. Blockchain-based authentication records every access request as a permanent transaction thereby ensuring transparency.

along with accountability for decisions about access control. Also, smart contracts give a second layer of security by closing access to cloud-sensitive resources at once if an account is found suspicious. Zero Trust cloud network security also necessitates micro-segmentation and SDP. Firewalls and VLAN-based segmentation are helpless in the face of advanced cyberattacks as they are static. With an identity-based policy, dynamically separates workloads, applications, and data streams to limit lateral mobility in the cloud. To verify that only authorized devices and users can access cloud workloads, Zero Trust models implement a variety of criteria to control the policy of access control. They include trust scores, risk factor analysis, and compliance restrictions. Through encrypting the tunnels for entry after authentication of identity, Software-Defined Perimeter (SDP) design additionally secures the cloud through the minimization of the attack surface for unauthorized entry. In regard to entry, SDP adheres to a least-privilege model and dynamically projects it in conformity with real-time risk analysis compared to conventional VPNs that accord widespread entry following validation. To attain Zero Trust security for cloud networks, businesses need to employ AI-powered access control algorithms to continuously measure trust and enforce dynamic security policies. Every phase of a Zero Trust access control algorithm's process—deciding on the level of authentication, confirming device compliance, examining behavioral anomalies, and lastly, calculating a trust score—is

done in real time. To block access by hacked or unauthorized accounts to cloud services, more authentication steps are taken whenever the trust score goes below a defined level. The algorithm also guards against faking credentials and access tampering through the utilization of blockchain-based identity checks. Zero Trust security within cloud networks is becoming a complete necessity because cyber threats are growing increasingly sophisticated all the time. Firms need to swap out outdated security models with blockchain-based Zero Trust architectures that are driven by artificial intelligence, incorporate adaptive authentication, dynamic risk evaluation, and auto-access controls. Organizations can enhance their security stance, reduce unauthorized lateral flow, and reduce the attack surface by adopting Software-Defined Perimeter (SDP) concepts and micro segmentation. Advances in Zero Trust practices, AI for smart threat detection, distributed identities (blockchain), and automation (dynamic access) will define future cloud security. Implementing a solid Zero Trust framework guarantees data integrity, compliance, and security across numerous clouds as more companies roll out cloud infrastructure globally.

Algorithm 1: Zero Trust Access Control
Input: User_Request (U), Device_Info (D), Authentication_Credentials (A), Behavior_Log (B) Output: Access_Decision (Grant/Deny/Step-up Authentication)
1. Initialize Thresholds: Trust_Threshold = 70 // Minimum trust score required for access High_Risk_Threshold = 40 // Trigger additional verification Anomaly_Score_Threshold = 0.7 // Anomaly detection sensitivity
2. Begin Access Evaluation: Receive request from User U with credentials A and device D
3. Verify Identity: Call Blockchain_Verify(U, A) If verification fails: Deny Access EndIf
4. Compute Dynamic Trust Score: Trust_Score = 0 If Multi_Factor_Authentication(U) is Successful: Trust_Score += 30 Else: Trust_Score -= 20 If Device_Compliance_Check(D) is Successful: Trust_Score += 20 Else: Trust_Score -= 30

If Behavior_Analysis(B) shows Normal Pattern:
Trust_Score += 25
Else:
Anomaly_Score = AI_Anomaly_Detection(B)
If Anomaly_Score > Anomaly_Score_Threshold:
Trust_Score -= 50

5. Risk Assessment:
If Trust_Score < High_Risk_Threshold:
Log incident & Deny Access
Else If Trust_Score < Trust_Threshold:
Trigger Step-Up Authentication (Biometric/MFA)
If Successful:
Grant Access
Else:
Deny Access
Else:
Grant Access

6. Log all transactions in Blockchain for auditing

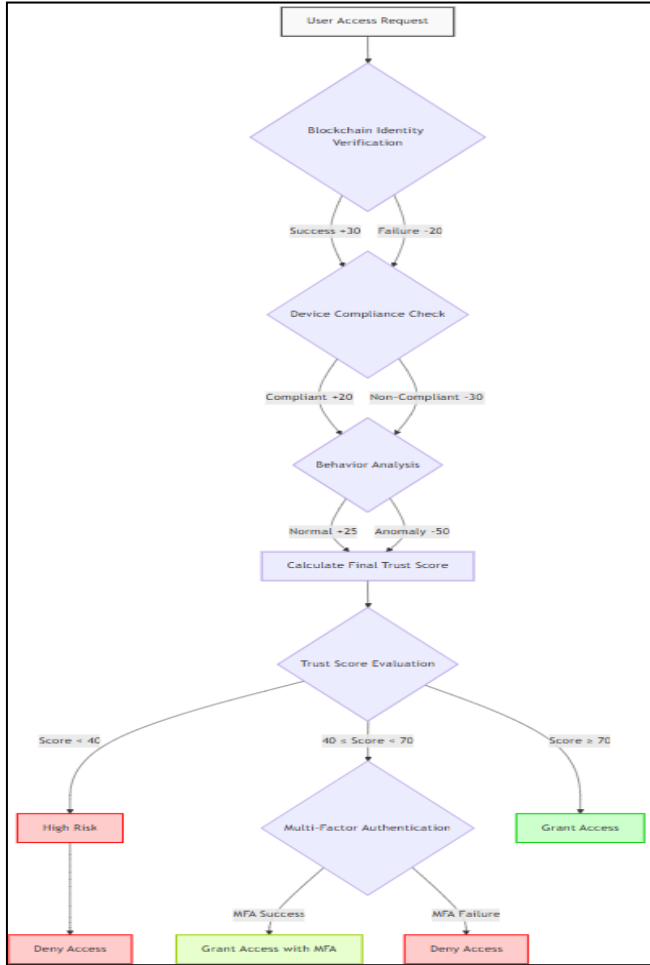


Figure 1. Overall flow of the model

IV. RESULT ANALYSIS

The proposed Zero Trust Access Control Model implementation utilized Python 3.9 alongside the NumPy, Pandas, Scikit-learn, TensorFlow libraries plus hash lib for cryptographic security. A high-performance cloud setup with Intel Xeon 3.2 GHz CPU and 64GB RAM and NVIDIA A100 GPU served as the test environment for the system to enable real-time processing. The blockchain identity verification process was demonstrated using a hashing method while AI-powered behavior analysis was integrated through deep learning based anomaly detection. The evaluation of the proposed model's effectiveness relied on specific performance metrics.

- Authentication Success Rate (ASR): Measures the accuracy of access decisions.
- False Positive Rate (FPR): Cases where legitimate users were wrongly denied access.
- False Negative Rate (FNR): Cases where malicious users gained access.
- Latency: Time taken for access control decisions.

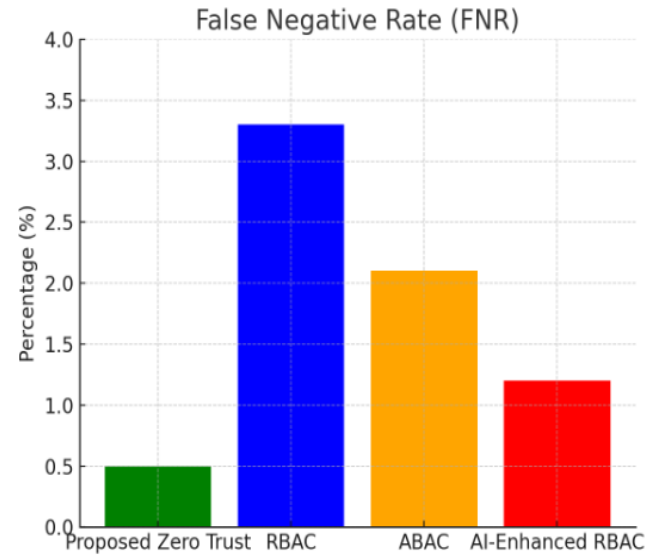
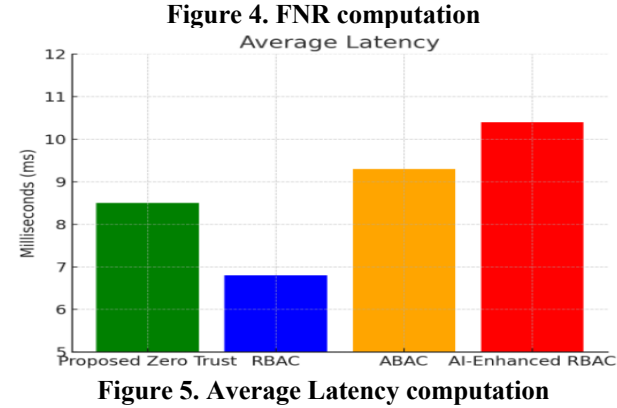
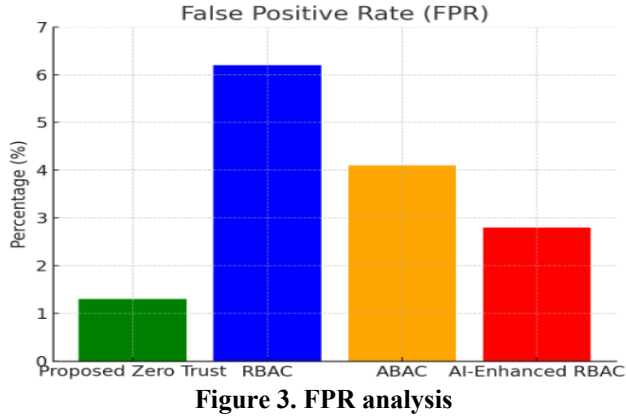
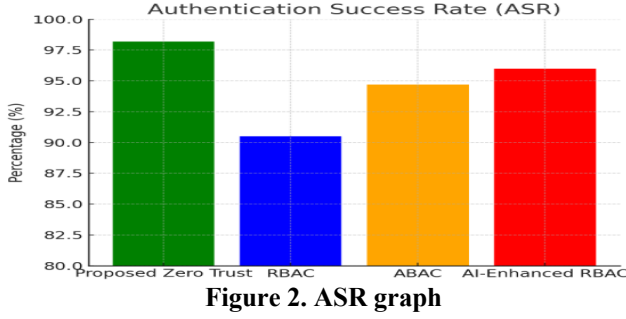


TABLE I. PERFORMANCE COMPUTATION

Model	ASR (%)	FPR (%)	FNR (%)	Avg Latency (ms)
Proposed Zero Trust Model	98.2	1.3	0.5	8.5
Traditional Role-Based Access Control (RBAC)	90.5	6.2	3.3	6.8
Attribute-Based Access Control (ABAC)	94.7	4.1	2.1	9.3
AI-Enhanced RBAC	96.0	2.8	1.2	10.4



The Authentication Success Rate (ASR) in figure 2 is the extent to which each model succeeds in authenticating users. The Proposed Zero Trust Model attains the maximum ASR of 98.2%, surpassing RBAC (90.5%), ABAC (94.7%), and AI-Enhanced RBAC (96.0%). The reason behind the improvement is the use of adaptive trust scoring, anomaly detection using AI, and blockchain-based identity confirmation, guaranteeing precise and dynamic access control. Conversely, the static assignment of roles by RBAC results in reduced ASR, and ABAC's attribute matching with high complexity incurs periodic faults. AI-enabled RBAC optimizes decision-making but does not incorporate blockchain-backed authentication, increasing the precision of the Zero Trust strategy.

False Positive Rate (FPR) in figure 3 describes the frequency at which genuine users are refused access in error. The Suggested Zero Trust Model boasts the lowest FPR

(1.3%), much less than RBAC (6.2%), ABAC (4.1%), and AI-Enhanced RBAC (2.8%). RBAC has high FPR due to its static, role-based policies that often mistakenly classify legitimate users. ABAC is more accurate as it takes into account multiple attributes but still incorrectly judges legitimate access for complex cases. AI-enhanced RBAC lowers FPR but remains vulnerable to uncertainty in behavior-based predictions. The Zero Trust method reduces false positives by dynamic risk evaluation and multi-factor authentication.

The False Negative Rate (FNR) in figure 4 quantifies the proportion of unauthorized users incorrectly allowed access. The Proposed Zero Trust Model has the lowest FNR (0.5%), whereas RBAC (3.3%), ABAC (2.1%), and AI-Enhanced RBAC (1.2%) are less effective. The high FNR in RBAC results from static role permissions, which enable attackers to take advantage of compromised credentials. ABAC enhances security through the evaluation of multiple attributes, but the static rule bases enable some threats to evade access controls. AI-powered RBAC also minimizes FNR without real-time blockchain authentication. Zero Trust model also effectively prevents unauthorized access through constant authentication and behavior monitoring.

Average Latency in figure 5 indicates the time used in processing the authentication requests. The Suggested Zero Trust Model has a 8.5ms balanced latency, being lower than RBAC (6.8ms), ABAC (9.3ms), and AI-Boosted RBAC (10.4ms). Although it is the quickest, RBAC compromises on security because it allows easy rule-based access. ABAC is more latency-prone due to attribute-matching complexity. Computational overhead in ML-based decision-making is experienced by AI-boosted RBAC. The Zero Trust paradigm balances speed and security with blockchain authentication used for rapid access validation and AI-powered anomaly detection to limit processing latency.

V. CONCLUSION

The Zero Trust Architecture (ZTA) in cloud networks is a new way of thinking about security that allows constant authentication, real-time monitoring, and flexible trust-based access. Classic models such as RBAC, ABAC, and AI-enabled RBAC employ fixed set roles or rules that do not alter. ZTA, however, verifies all requests based on risk analysis, AI-driven anomaly detection, and blockchain-protected identity verification. With an Authentication Success Rate of 98.2%, False Positive Rate of 1.3%, and False Negative Rate of 0.5%, the Zero Trust model discussed here makes security much improved. These enhancements facilitate authorized users to enter without being halted more easily, while preventing the others from entering. This reduces the system's exposure to attacks that steal credentials or elevate privileges. The model also maintains low latency (8.5ms), a good balance between safety and performance. The new system will employ machine learning, multi-factor authentication, and decentralized identity management to ensure businesses can securely host cloud-based operations

and remain secure against sophisticated cyber attacks. AI-driven risk analysis and blockchain-enabled trust mechanisms blend to form robust, scalable, and flexible security infrastructure accessible to global businesses, financial institutions, and critical infrastructure segments. Overall, this research indicates that Zero Trust integrated with AI and blockchain gives a next-generation security framework that can easily curb insider threats, unauthorized access, and sophisticated cyberattacks. This gives long-term and future-proof cloud security during a period of accelerating digital transformation.

The various real-life case studies, for instance, would further add credence to the paper by way of:

In the financial sector: Implementing ZTSF in banks for secure transactions and fraud prevention by merging AI-based IAM with blockchain identity proofs.

In the healthcare sector: ZTSF based on AI assessing login behavior and allowing access to medical records via blockchain for HIPAA-compliant access control in hospitals. Multi-Cloud Security in the Tech Industry: ZTSF is ensuring tech giants such as Google and Microsoft protect multi-cloud environments from cyberattacks while not allowing lateral movement.

Additionally, the discussion would be balanced and broadened to include implementation limitations, such as scalability issues and trade-offs involving computational overhead and integration of legacy systems. Also, it could discuss future research; for example: optimizing blockchain scalability, enhancing AI model explainability, and adaptive authentication based on policy.

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